Embedded Systems Development

Exercise 1

1. Construct a microcontroller-based system to chase five LEDs from left to right repeatedly.

   a. If the system is supplied with 5 V and an LED is driven with 2 V and 15 mA, what is the resistor required to limit the current to the LED? Show your workings clearly.

   ……

   According to Ohm's Law $R = \frac{V_S - V_L}{I_L}$ where V_S is supply voltage, V_L is voltage across LED, I_L is current through LED and R is the resistor required.

   ……

   $$R = \frac{5V - 2V}{0.015A} = 200\Omega \approx 220\Omega$$

   ……

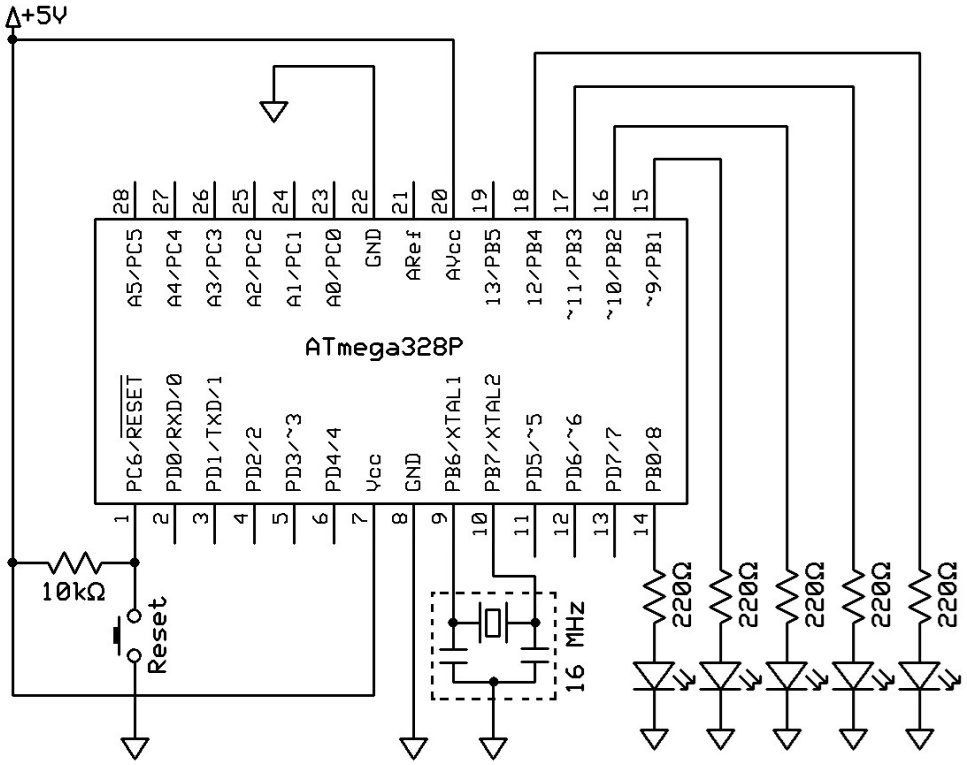
   b. Prepare a Bill of Materials (BOM).

   ……

   1 × Arduino Uno
   1 × Breadboard
   5 × LED
   5 × 220Ω Resistor
   Male to male breadboard jumper wires

   ……

   c. Complete the following schematic diagram for your hardware design.



d. Write the required firmware using the Arduino wrapper functions.

```
1   void setup()
2   {
3     for (unsigned char i = 8; i <= 12; i++)
4       pinMode(i, OUTPUT);
5   }
6   void loop()
7   {
8     for (unsigned char i = 8; i <= 12; i++)
9     {
10      digitalWrite(i, HIGH);
11      delay(200);
12      digitalWrite(i, LOW);
13    }
14  }
```

920 bytes

e. Write the required firmware in C.

```c
#include <avr/io.h>
#define F_CPU 16000000UL
#include <util/delay.h>
int main(void)
{
    DDRB = 0b00011111; // configure PB0-PB4 as outputs
    while (1)
    {
        for (unsigned char i = 0; i <= 4; i++)
        {
            PORTB |= (1 << i); // set bit i and make PBi high
            _delay_ms(200);
            PORTB &= ~(1 << i); // clear bit i and make PBi low
        }
    }
}
```

200 bytes (78% improvement)

```c
#include <avr/io.h>
#define F_CPU 16000000UL
#include <util/delay.h>
int main(void)
{
    DDRB = 0b00011111; // configure PB0-PB4 as outputs
    while (1)
    {
        PORTB = 0b00000001;
        while(!(PORTB & 0b00100000)) // repeat until bit 5 is set
        {
            _delay_ms(200);
            PORTB <<= 1; // shift bits 1 position to left
        }
    }
}
```

172 bytes (further 3% improvement)

2. Modify your system to drive the five LEDs in Knight Rider pattern.
   a. Write the required firmware using the Arduino wrapper functions.

```
1   void setup()
2   {
3     for (unsigned char i = 8; i <= 12; i++)
4       pinMode(i, OUTPUT);
5   }
6   void loop()
7   {
8     for (unsigned char i = 8; i <= 12; i++)
9     { // left to right
10      digitalWrite(i, HIGH);
11      delay(200);
12      digitalWrite(i, LOW);
13    }
14    for (unsigned char i = 11; i >= 9; i--)
15    { // right to left
16      digitalWrite(i, HIGH);
17      delay(200);
18      digitalWrite(i, LOW);
19    }
20  }
```

986 bytes

b. Write the required firmware in C.

```c
#include <avr/io.h>
#define F_CPU 16000000UL
#include <util/delay.h>
int main(void)
{
    DDRB = 0b00011111; // configure PB0-PB4 as outputs
    while (1)
    {
        for (unsigned char i = 0; i <= 4; i++)
        { // left to right
            PORTB |= (1 << i); // set bit i and make PBi high
            _delay_ms(200);
            PORTB &= ~(1 << i); // clear bit i and make PBi low
        }
        for (unsigned char i = 3; i >= 1; i--)
        { // right to left
            PORTB |= (1 << i); // set bit i and make PBi high
            _delay_ms(200);
            PORTB &= ~(1 << i); // clear bit i and make PBi low
        }
    }
}
```

260 bytes (74% improvement)

```c
#include <avr/io.h>
#define F_CPU 16000000UL
#include <util/delay.h>
int main(void)
{
    DDRB = 0b00011111; // configure PB0-PB4 as outputs
    PORTB = 0b00000001; // set bit 0 and make PB0 high
    while (1)
    {
        while (!(PORTB & 0b00010000)) // repeat until bit 4 is set
        { // left to right
            _delay_ms(200);
            PORTB <<= 1; // shift bits 1 position to left
        }
        while (!(PORTB & 0b00000001)) // repeat until bit 0 is set
        { // right to left
            _delay_ms(200);
            PORTB >>= 1; // shift bits 1 position to right
        }
    }
}
```

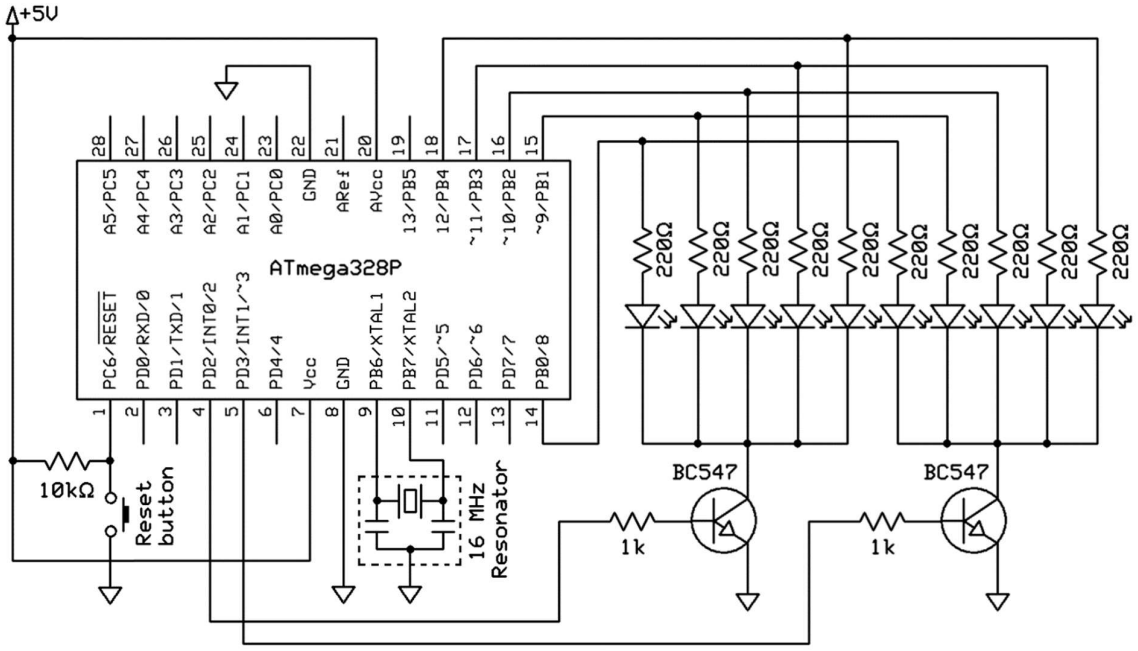
202 bytes (further 6% improvement)

3. Extend the system developed in Question 2 to drive 10 LEDs using a maximum of seven I/O pins.

   Hint: Use two BC547 transistors with 1 kΩ base resistors to switch the GND connection between the cathodes of the first and second sets of five LEDs.



   a. Complete the following schematic diagram for your hardware design.



b. Write the required firmware in C.

```c
#include <avr/io.h>
#define F_CPU 16000000UL
#include <util/delay.h>
void chaseRight(void)
{
    while (!(PORTB & 0b00010000)) // repeat until bit 4 is set
    {
        _delay_ms(200);
        PORTB <<= 1; // shift bits 1 position to left
    }
}
void chaseLeft(void)
{
    while (!(PORTB & 0b00000001)) // repeat until bit 0 is set
    {
        _delay_ms(200);
        PORTB >>= 1; // shift bits 1 position to right
    }
}
int main(void)
{
    DDRB = 0b00011111; // configure PB0-PB4 as outputs
    DDRD = 0b00001100; // configure PD2 & PD3 as outputs
    PORTD = 0b00000100; // enable first set of LEDs
    PORTB = 0b00000001; // turn on first LED
    while (1)
    {
        chaseRight(); // chase first set of LEDs to right
        _delay_ms(200); // wait for fifth LED
        PORTD = 0b00001000; // enable second set of LEDs
        PORTB = 0b00000001; // turn on sixth LED
        chaseRight(); // chase second set of LEDs to right
        chaseLeft(); // chase second set of LEDs to left
        _delay_ms(200); // wait for sixth LED
        PORTD = 0b00000100; // enable first set of LEDs
        PORTB = 0b00010000; // turn on fifth LED
        chaseLeft(); // chase first set of LEDs to left
    }
}
```